Estimating the Probability of Winning a VALORANT Match Using Historical Player Performance

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Can Historical Player Performance Predict Victory?

PAST MATCHES → FUTURE OUTCOME

Estimating match win probability from recent player statistics

01

TEAM-BASED GAME

Players work together, but individual stats do not tell the whole story.

02

HISTORICAL SIGNALS

Past match statistics provide measurable information about recent performance.

03

RESEARCH QUESTION

Can past performance help estimate the chance of winning a future match?

Predicting VALORANT Match Win Probability

Input: average statistics from the previous five matches

ACS — **Average Combat Score**
Overall combat performance

K/D — **Kill-to-Death Ratio**
Kills compared with deaths

KAST% — **Kill / Assist / Survive / Trade**
Share of rounds with one of these outcomes

HS% — **Headshot Percentage**
Share of hits that are headshots

DDΔ / Round — **Damage Difference per Round**
Damage dealt minus damage received

Win Rate — **Previous-match win rate**
Percentage of previous matches won

TARGET (Y) 1 = Win · 0 = Loss

OUTPUT Estimated probability of winning

Only previous-match statistics are used. The target match's own statistics are excluded.

Data Collection and Overview

Current-season Competitive match statistics from VALORANT Tracker

40
PLAYERS

400
PLAYER-MATCH RECORDS

208 / 192
WINS / LOSSES

DATA COLLECTION
Recorded each player's match result and performance metrics.

EXAMPLE OF OUR DATASET

1	playerID	WINRATE %	KD	ACS	KAST %	DDΔ/rnd	HS %
2	ying#0826	66.7	1.46	258.7	78.6	47	21.1

playerID	matchID	WIN OR LOSE(1	ACS	KD	DDΔ/Round	HS %	KAST %
ying#0826	1	1	266	3	116	18	93
	2	0	258	1.4	39	30	80
	3	0	327	1.7	65	33	73
	4	1	211	1.4	22	26	86
	5	1	232	1.7	48	18	91

Source: VALORANT Tracker · Records are player-match observations, not necessarily unique actual matches.

What Previous Esports Prediction Research Shows

Different approaches to predicting competitive outcomes

01 — PLAYER PERFORMANCE
- League of Legends studies use player performance metrics
- Historical statistics may contain predictive signal

02 — TEAM INTERACTIONS
- Team outcomes depend on multiple players
- Individual statistics may not capture the full match context

03 — VALORANT RESEARCH
- Existing work has used tactical and in-match features
- Our project instead uses only historical player performance

Our Focus:
Can recent player-level statistics help estimate the next-match win probability?

Challenges and Limitations in Our Dataset

What our data captures — and what it does not

SAMPLE SIZE

40–45 players
10 matches per player
Limited representation of the full VALORANT population

WITHIN-PLAYER DEPENDENCE

Each player contributes multiple prediction observations
Rolling windows overlap across predictions
Observations from the same player may be correlated

MISSING MATCH CONTEXT

Not included:
Teammates / opponents
Team composition
Maps / agents

AVOIDING INFORMATION LEAKAGE

Only statistics from matches before the target match will be used
Overall player statistics will mainly be used descriptively

From Match History to Prediction

Using a five-match rolling window

eg.

ying#0826	1	1	266	3	116	18	93
	2	0	258	1.4	39	30	80
	3	0	327	1.7	65	33	73
	4	1	211	1.4	22	26	86
	5	1	232	1.7	48	18	91
	6	?	?	?	?	?	?

ROLLING PREDICTION RULE

- Previous 5 matches → predict the next match
- Example: Matches 1–5 → Match 6
- Target-match statistics are excluded

HISTORICAL FEATURES

- Average ACS
- K/D
- KAST%
- HS%
- DDΔ / Round
- Recent Win Rate

Baselines and Predictive Models

Do more complex models improve prediction?

NAIVE BASELINE

Predict the same 0.50 win probability for every match

HISTORICAL BASELINE

Recent Win Rate
Use the player's win rate from the previous five matches

LOGISTIC REGRESSION

Interpretable Probability Model
Estimates win probability directly
Measures relationships between historic
features and outcome

DECISION TREE

Nonlinear Model
Can capture nonlinear patterns
Can model interactions between performance statistics

EVALUATION

Primary: Log Loss
Secondary: Accuracy
Optional: ROC-AUC

Conclusion

What We Aim to Learn

- Use recent player performance to estimate next-match win probability
- Compare simple baselines with Logistic Regression and Decision Tree
- Avoid future information leakage with a rolling five-match window
- Evaluate whether historical statistics provide meaningful predictive value

The key question is whether recent player performance adds predictive value beyond simple baselines.

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